

YOLO Detection ONNX RUNTIME Model Inference Report

Backend Engine: ONNX Runtime

Phase 1: Configuration & System Metadata

Model Name	yolo26n.onnx
Input Dimensions	640x640
Source Images Directory	/workspace/Images
Processed Images Output	/workspace/ONNX_Output
OS	Ubuntu 24.04.4 LTS
Python Version	3.12.3
PyTorch Version	2.13.0+cu130
Ultralytics Version	8.4.75
Inference Device	NVIDIA RTX A4000
CUDA Version	13.0
Model Architecture	YOLO (Detect)
Total Parameters	2,408,981
File Size	9.86 MB
Number of Classes	80
Precision Mode	FP32

Model Trained Class Names:

person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, traffic light, fire hydrant, stop sign, parking meter, bench, bird, cat, dog, horse, sheep, cow, elephant, bear, zebra, giraffe, backpack, umbrella, handbag, tie, suitcase, frisbee, skis, snowboard, sports ball, kite, baseball bat, baseball glove, skateboard, surfboard, tennis racket, bottle, wine glass, cup, fork, knife, spoon, bowl, banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake, chair, couch, potted plant, bed, dining table, toilet, tv, laptop, mouse, remote, keyboard, cell phone, microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush

Phase 2: Inference Performance Metrics

Total Images Processed	5
Total Pipeline Time	2293.57 ms
Total Inference Time in ms	1659.88ms
Preprocess Latency	14.02 ms per image
Inference Latency	331.98 ms per image
Postprocess Latency	112.72 ms per image
Throughput	3.01 Images/sec
Peak Memory Usage	454.00 MB

Phase 3: Detailed Predictions Summary

Image File	Detected Objects (Confidence)	Bounding Boxes (xyxy)
Cat_November_2010-1a.jpg.webp	cat (92.5%)	[207, 142, 891, 1214]
images.jpeg	dog (94.3%)	[87, 148, 422, 514]
images1.jpeg	zebra (92.7%)	[30, 173, 587, 415]
istockphoto-1189925691-612x612.jpg	bus (85.5%)	[69, 81, 583, 350]
tigermain.jpg	zebra (93.8%)	[672, 421, 1875, 1090]